

Sukabumi station outage, 28 August – 1 September 2026

The station was unreachable for 4.8 days. It was running throughout, on schedule and at a normal duty cycle; the cause was an unfunded prepaid SIM. This report sets out what the station's own power log measures, and which earlier conclusions it revises.

Site Sukabumi (pilot) Silent 28 Aug 06:30 WIB → 1 Sep 18:00 UTC
Evidence /var/log/wp5d.log · 3.0 MB · 5,714 boots Issued 1 Sep 2026

0.87× DUTY CYCLE DURING THE OUTAGE, AGAINST THE WEEK BEFORE IT	48/day SCHEDULED BOOTS, UNBROKEN BEFORE AND THROUGH	5.15 min LONGEST SINGLE WAKE IN THE OUTAGE; EARLIER ANALYSIS STATED 25
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WHAT HAPPENED

The uplink failed, not the station

On 28 August at 06:30 WIB, Sukabumi stopped reaching the server. Nothing arrived for 4.8 days. At 18:00 UTC on 1 September it returned and flushed roughly 4,500 buffered rows in two minutes — the sensor readings it had been taking, and storing, throughout.

The cause was the **Telkomsel prepaid account running out of money**. No balance, no data. The modem stays registered and healthy the whole time, which is precisely the signature we saw and could not explain: a station visible to the Tailscale control plane, occasionally answering on port 22, and never delivering anything.

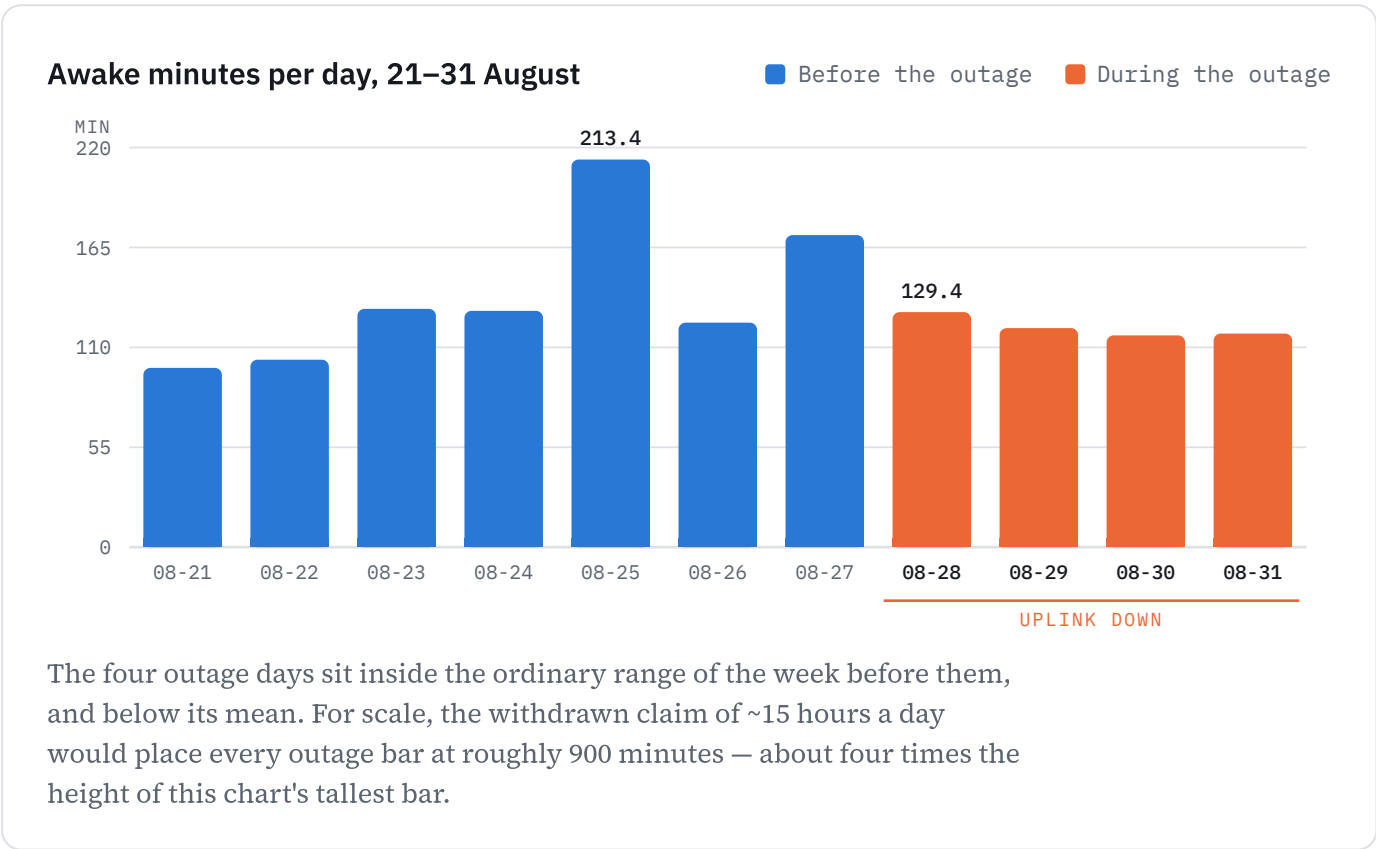
One unpaid bill produced the entire event. Nothing about the hardware failed.

THE CORRECTION

Duty cycle during the outage was normal

The first analysis of the returned data concluded the station had been awake roughly 15 hours a day instead of 1.6, running nearly every cycle to the Witty Pi's 25-minute backstop at about nine times its energy budget. That reading came from wake-minutes *inferred* from the spacing of sensor rows.

The Witty Pi keeps its own log of when it powered the Pi. We pulled the whole file — 5,714 boots back to 7 April — and it measures the duty cycle directly. Each cycle records `Startup reason` after the clock is set, and `Exit now.` at shutdown. The difference is the awake time.



MEASURED FROM /VAR/LOG/WP5D.LOG · OUTAGE DAYS SHADED

DAY	CYCLES	MEDIAN WAKE	LONGEST WAKE	AWAKE/DAY
08-21	47	2.03 min	4.23 min	98.7 min
08-22	48	2.05 min	4.37 min	103.2 min
08-23	48	2.74 min	5.22 min	131.2 min
08-24	48	2.73 min	5.18 min	130.1 min
08-25	45	2.92 min	23.98 min	213.4 min

DAY	CYCLES	MEDIAN WAKE	LONGEST WAKE	AWAKE/DAY
08-26	48	2.23 min	5.18 min	123.6 min
08-27	47	2.73 min	24.27 min	171.8 min
08-28	48	2.80 min	5.15 min	129.4 min
08-29	48	2.73 min	2.95 min	120.6 min
08-30	48	2.39 min	2.95 min	116.6 min
08-31	48	2.42 min	2.92 min	117.6 min

What the measurements rule out

Across the whole deployment — 5,244 measured cycles since April — the median wake is 1.95 minutes and only **eight cycles have ever exceeded 20 minutes**. Two of those eight fall in this table, on 25 and 27 August. Both are *before* the outage began.

During the outage itself the longest single wake was 5.15 minutes, and on the final three days nothing exceeded 2.95 minutes. The station did not approach the 25-minute backstop, and there is no drain signature in this window.

The choice of baseline is what produced the original error. Against the quiet week of 10–13 August (94–101 min/day) the outage looks like a mild 1.25× rise; against the week immediately before it (mean 139 min/day) it is 0.87× — slightly less awake than normal. Daily awake time drifts between roughly 95 and 215 minutes as a matter of course, and the outage days land in the middle of that band.

CLAIM LEDGER

Status of each conclusion

Each claim is marked by how it is known. Measured claims are taken from the station's own power log; claims previously inferred from server-side row spacing have been re-examined against it.

● REPORTED

The uplink failed because the Telkomsel prepaid account ran out of money.

Source: Tom, 1 Sep. Consistent with every observed symptom; not independently verified from the station.

● MEASURED	The station ran 48 scheduled boots a day, unbroken, before and through the outage. wp5d.log · 5,714 boots · 5,645 "Scheduled Startup"
● MEASURED	Duty cycle during the outage was normal — 117–129 min/day against a preceding-week mean of 139. wp5d.log · 5,244 timed cycles
● MEASURED	V-IN never fell below 12.149 V across the outage, and reached 13.71 V. 679 samples in the flushed backlog, against 11 for the whole project before it
● WITHDRAWN	"Awake roughly 15 hours a day instead of 1.6, at about 9× its energy budget." Inferred from sensor-row spacing. The Witty Pi log measures ~2 h/day.
● WITHDRAWN	"ORC-OS never completed its task, pinning the Pi to the 25-minute backstop." The task completed every cycle in under three minutes. There was no hang, and the parallel drawn to the maintenance-mode chain does not apply.
● WITHDRAWN	The night-onset pattern, the dawn windows, the three-missed-dawns argument, and the case for a site visit. All reasoned from absence of rows on a station that was working. A visit would have found nothing wrong.
● WITHDRAWN	The recovery-voltage latch theory. 13.0 V is reached routinely. Every V-IN sample the project held before this flush came from a 3-hour pre-dawn window, which is why the pack looked incapable of it.
○ WEAKENED	"The pack is not the fault." Still true for this event — V-IN held. But it rested on surviving 9× duty, and the real figure was 0.87×. The pack was never tested under load here.

Tooling and operating policy

Detection

Nothing noticed for 4.8 days. The outage began as a single missed upload and no alarm existed for that. `db_watch.py` now alerts when the newest sensor row passes 95 minutes old — three missed cycles — and again when uploads resume, reporting how many rows arrived in the flush. That closes an alerting gap open since July, at least for the station's own uplink.

Collection

The recovery watcher grabbed only `tail -n 400` of `wp5d.log`, sized for a seconds-long wake window; it covered the last 20 hours of a 4.8-day outage. It now takes the whole file as a second grab, gzipped over the wire, only after the tail is safely down. That is the artifact this report is built on.

That change had a defect on first use: it re-grabbed 3 MB on every pounce, and the watcher re-pounces for as long as port 22 stays open. Three identical copies in 90 seconds — 9 MB of prepaid LTE for 3 MB of information, on the SIM whose running dry caused the outage. A 24-hour guard is now in place and the duplicates deleted.

Unattended operation — stopped

The watchers are not passive: one opens an SSH session to the station and pulls files. They now run only for the lifetime of an active session. The systemd unit that started them at every login has been stopped, disabled and retired, with its reasoning rewritten in place so it is not simply re-created; `enable-linger` was never on and stays off.

FOUND WHILE APPLYING THIS

A session left running since 28 August had been watching the station unattended for four days. It re-enabled the retired unit 82 seconds after it was disabled, killed the replacement watcher, and respawned watchers from a scratchpad belonging to a session that had already ended. Its output went nowhere anyone could see — which is why the watchers looked dead when this work began.

The cost of the policy is real and accepted: wakes that land between sessions are missed.

WHERE WE STAND

Open items, in order of risk

- 01 **Nothing watches the SIM balance.** This is the actual root cause and it has no fix. A prepaid account that silently empties is a single point of failure for the whole station — not instrumented, not alerted, not on anyone's calendar. It will do this again.
- 02 **No connectivity-loss handling in software.** Not the cause of the drain we thought we saw, but still absent: nothing treats loss of uplink as a terminating condition rather than something to retry against.
- 03 **The station's power behaviour is now measurable but unmeasured.** The flush delivered 679 V-IN samples against 11 for the whole project before it, and only because a backlog forced them through. Routine V/I telemetry is deployed but the record is still thin.
- 04 `station_gaps.py` **rests on a false premise.** Absence of rows does not record downtime — that failed completely here. Any analysis built on it needs re-checking against the Witty Pi log.

Current state

- Station is on its normal 30-minute cadence. Newest row 09-02 01:00:27 WIB.
- Both watches armed, session-scoped: the wake-window grabber and the upload-stall alarm.
- Full wp5d.log archived — 5,714 boots, 7 April to 1 September.
- Uncommitted: the duty-cycle correction in the issue log, and the 24-hour grab guard.

Sukabumi is a pilot. Data loss there is not an emergency; the findings are what matter. All figures measured from the station's own Witty Pi log unless marked otherwise. Awake time is taken to daemon exit rather than to the power rail dropping, so every figure is a slight underestimate — equally on both sides of the comparison the conclusion rests on.